MILITARY SPECIFICATION

LATHES, BENCH, VARIABLE SPEED, PLAIN AND TURRET, 9 INCH

This specification is approved for use by all Departments and Agencies of the Deparment of Defense.

1. SCOPE

1.1 Scope. This specification covers cabinet mounted, metal cutting bench lathes having a variable, high speed headstock and manually operated tool slides.

1.2 Classification. The lathes covered by this specification are of the following types. The type required for a specific procurement shall be as specified (see 6.2.1).

Type I - Plain bench lathe
TypeII- Turret bench lathe

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center Memphis, Tennessee 38114, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standard for Federal Services

(Copies of standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting officer.)

2.1.2 Other government documents, drawings and publications. The following other government documents, drawings, and publications form a part of this specification to the extent specified herein.

u. s. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910)

(Application for copies should be addressed to the Superintendent of Documents , Government Printing Officer, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issue of the document which is indicated as DoD adopted shall be the issue listed in the current DoDISS and supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B5.8 - Chucks and Chuck Jaws
B11. 6 - Safety Requirements for the Construction, Care and Use of Lathes
ANSI/IEEE 268 - Metric Practice

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-D-3951 - commercial Packaging, Standard Practice for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA NO. 79 - Electrical Standard -f or Metalworking Machine Tools and Plastics Machinery (ANSI C113.1)

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the test of this specification and reference cited herein, the test of this specification shall take precedence.

3* REQUIREMENTS

3.1 First article. When specified (see 6.2.1) , the contractor shall furnish a sample unit for first article inspection and approval (see 4.2 and 6.3).

3.2 Design. The lathe shall be new and one of the manufacturer's current models capable of conforming to the accuracy requirements for a bench lathe as specified herein. The lathe shall be capable of performing turning, boring, facing, and threading operations within the capacities of the size specified. The machine shall include all components parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjusment and repair.

3.2.1 Measurement Systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the machine. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2. 1). In this specification, all measurements, dimensions, sizes and capacities are given in the U.S. Customary System of Units (US). These measurements may be converted to the International System of Units (SI) through the use of the conversion factors and methods specified in ANSI/IEEE 268.

3.2.1.1 Measuring and indicating device graduations. When specified (see 6.2.1), measuring and indicating devices such as scales, carriage stops, dial indicators, pressure gages, temperature indicators and all other similar devices shall be graduated in the specified system (US or SI) of measurements l Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system

3.2.1.2 Dual graduations. When specified (see 6.2.1) measuring and indicating devices shall be graduated in both the US and the ST System of Measurements. When a dual US and SI System is furnished, feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

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3.2.3 Controls. All operating controls shall be located convenient to the operator at his normal work station.

3.2.4 Safety and health requirements. Covers, quads, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance and repair of the parts. All machine parts, components, mechanisms and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the machine itself. In addition, the machine shall comply with all requirements of ANSI B11.6 that are designated therein as the responsibility of the machine manufacturer. In the event of conflict between the requirements of the OSHA and ANSI Standards, the requirements of OSHA shall apply. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

3.2.5 Lubrication. Means shall be provided to insure adequate lubrication for all roving parts. Each recirculating system shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.6 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of scale and mismatching. No process such as welding, peening, or filling with solder or paste shall be used for reclaiming any defective parts.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts and other fasteners shall be installed in a manner that prevents change of tightness. Those subject to removal or ad justment shall not be swaged, peened, staked or otherwise permanently installed.

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3.3.4 Surfaces. All surfaces shall be cleaned and free of sand, dirt, fins, sprues, flash, scale, flux and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1) , the machine shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Dials. All dials that indicate stock removal or tool movement shall be graduated in increments not larger than 0.001 inch (US) or .02mm (SI). Dial diameters shall be such that graduations may be read from the normal operating position. Dials shall be permanently and legibly engraved or etched on a non-glare background.

3.3.8 Plates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background.

3.3.9 Gears. All gears shall be machined from a material suitable for the intended purpose. All headstock spindle drive train gears shall be heat treated by a process that will impart the hardness and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure or premature wear.

3.4 Components, Unless otherwise specified in paragraph 3.11, the type I and type II lathes shall conform to the following requirements.

3.4.1 Bed. The lathe bed shall be a ferrous casting, treated during the manufacturing process to retain the lathe accuracies specified in table II. The bed shall provide the rigidity required to maintain alignment of the headstock and tool carrier under all cutting conditions. The bed ways shall extend from the headstock through the end of the bed to facilitate interchanging components. All surfaces of the ways on which the tool slides bear for support or guidance shall be hardened and ground. Cast ways shall have a hardness of not less than Rockwell C50. Steel insert type ways shall have a hardness of not less than Rockwell C60.

3.4.2 Base. The lathe base shall be of the cabinet type, fabricated of castings or steel weldments. It shall be ribbed or braced to provide a firm support for the lathe. The top of the base shall form an oil tight chip and coolant pan and shall have a screened reservoir for collecting coolant. The headstock drive motor and variable speed unit shall be housed within the base. Space shall be provided for storage of lathe accessories. Suitable means shall be provided for securing and leveling the base on a foundation.

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3.4.3 Headstock. The headstock casting shall enclose the spindle and spindle drive mechanism. The drive mechanism shall be arranged to permit changing endless drive belts without disassembling the spindle or disturbing the spindle bearing adjusment.

3.4.4 Headstock spindle. The spindle shall be a draw-in collet type, the collets being seated against the inner taper surface of the spindle nose. The outer surface of the spindle nose shall be of the taper, key drive design for mounting chucks and face plates. It shall permit operation of the spindle in both directions without loosening of the attachments. The inner and outer surfaces of the spindle nose shall have a hardness of not less than Rockwell C60. The spindle assembly shall be in dynamic balance for vibrationless operation throughout the range of speeds. It shall be mounted in preloaded, lubricated and sealed-for-life bearings. The rear end of the spindle shall be fitted with a handwheel for rotating the spindle manually. when specified (see 6.2.1), the spindle nose shall be of the threaded type for mounting chucks and face plates or stationary collets.

3.4.5 Spindle drive. The drive mechanism shall operate the spindle in both the forward and reverse directions of rotation through multiple v-belts. The drive mechanism shall provide infinitely variable spindle speeds, The selectd speeds shall remain constant regardless of variations in the cutting load. Spindle speeds shall be selected by a single speed control and the chosen speed shall be indicated in revolutions per minute. Quick, shockless spindle braking shall be provided. Means for adjustment of all drive belts shall be provided.

3.4.6 Type I lathe. The compnents in 3.4.6.1 thru 3.4.6.3 shall be provided with each type I lathe.

3.4.6.1 Compound slide rest. The compound rest shall consist of a base and bottom slide, swiveling top slide and feed screws. The base shall have a sliding fit on the lathe bed, be securely clamped in the work position and be readily removed from the bed. The bottom slide shall provide the transverse feed movements. The top slide shall be easily positioned to any angle in 360 degrees providing angular feed movements. The swivel shall be graduated in degrees with permanent, easily read markings to facilitate accurate angular setting of the top slide. A standard T-slot in the top slide shall permit mounting and positioning of interchangeable tool posts. The feed screws shall be totally enclosed and fitted with ball cranks and easy to read repositioning type fed dials. Full length gibs shall provide adjustment for wear between the slides.

3.4.6.2 Indexing square turret. The turret shall be for mounting in the T-slot of the compound rest. The turret shall hold 4 standard tools or holders, not less than 3/8 inch size. The entire indexing function (unlock, index, lock) shall be accomplished by action of a single index lever.

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3.4.6.3 Tailstock. The tailstock shall have a sliding fit on the bed ways and shall be securely clampable in the work position. The spindle shall have full bearing in the housing for support during drilling operations. Graduation marks on the spindle in increments not greater than 1/8 inch and a dial on the fed screw graduated to 0.001 inch shall measure feed movement. Tools shall be ejected from the spindle when fully retracted. The feed screw shall be mounted in ball thrust bearings for ease of operation.

3.4.7 Type II lathe. The components in 3.4.7.1 thru 3.4.7.4 shall be provided with each type II lathe.

3.4.7.1 Indexing turret. The turret shall be of the automatic clockwise indexing type, and index by hand in both directions, with six equally spaced tool stations. The unit shall consist of a base, slide and turret. The base shall have a sliding fit on the bed ways and shall be securely clamped in the work position and be readily removed f from the bed. The slide shall provide the longitudinal feed movement, operated by a hand lever, Adjustable gibs having a hardness not less than Rockwell C50 shall provide adjustment for wear. The turret shall rotate on preloaded ball or roller bearings. Station to station indexing functions shall be by an extreme rearward movement of the hand lever. The turret shall be securely locked in each work position after indexing. Each tool station shall have an adjustable stop which shall automatically index with the turret in both directions. Straight shank tool holders of the size required in table I shall be locked in the turret by suitable binders clamping on the shank.

3.4.7.2 Double tool cross slide. The cross slide shall be mounted on the lathe bed providing the transverse feed movements. It shall be easily moved to and securely clamped in any working position along the bed. Feed movements shall be by a hand lever, adjusted to provide a sensitive feel of the slide movement. The top of the slide shall have a standard T-slot, machined the full length for mounting and positioning tool holder blocks at the front and rear positions. Two blocks shall be furnished, having a standard T-slot machined 90 degrees to the slide slot for mounting and positioning tool holders. Adjustable gibs having a hardness not less than Rockwell C50 shall provide adjustment for wear between the slide and base.

3.4.7.3 Tool posts, cross slide. One each, front and rear mounting, open side, wedge type tool posts and one rear mountig cut-off tool holder shall be furnished. The tool posts shall accept not less than a 3/8 inch square tool bit. The cut-off tool holder shall accept 1/16, 3/32 and 1/8 inch wide cut-off blades.

3.4.7.4 coolant system. The coolant system shall include a sump or reservoir, a power driven prep, and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant.

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3.4.8 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to all requirements in this paragraph. The system shall conform to the requirements of ANSI/NFPA No. 79. The machine shall draw all of its electrical power f ran a single 230/460 volt, 3 phase, 60 Hz circuit, The machine shall be initially wired for operation on 460 volts. An identified terminal for grounding the machine when installed shall be mounted in or near the disconnect switch.

3.4.9 Motors. Motors shall have ball or roller bearings. Continuous duty Power ratings shall conform to the requirements of table I. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a dripproof enclosure.

3.5 Sizes and capacities. The sizes and capacities of the machine shall meet the requirements of table I as applicable for the machine type specif ied in the contract or order.

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TABLE I

SIZES AND CAPACITIES
(Dimensions in Inches)

CHARACTERISTIC	REQUIREMENT	
	Type I Plain Lathe	Type II Turret Lathe
General		
Swing over bed	9	9
Swing over bottom slide, compound rest	5-3/4	-
Swing over double tool slide	-	4-5/8
Distance between centers, tailstock flush with end of bed	16	-
Distance between face of spindle and face of turret	-	15-1/2
Headstock		
Spindle speed range, variable, RPM	230 to 3500	230 to 3500
Optional speed range	150 to 3500	150 to 3500
Draw-in collet capacity		
Round	1/8 to 1-1/16	1/8 to 1-1/16
Square	1/4 to 3/4	1/4 to 3/4
Hexagon	1/4 to 7/8	1/4 to 7/8
Optional, stationary collet capacity		
Round	-	1/8 to 1-1/16
Square	-	1/4 to 3/4
Hexagon	-	1/4 to 7/8
Diameter bar stock spindle will accept	1-1/16	1-1/16
Tailstock		
Spindle travel	3-1/2	-
Spindle taper, Morse taper	#2	-
Spindle diameter	1-1/4	-
Six position, hexagon or round tool turret		
Working stroke	-	3-1/4
Tool shank size, diameter	-	5/8
Compound rest		
Top slide travel	4-1/4	-
Bottom slide travel	5	-
Double tool cross slide, working stroke	-	3-3/4
Motor horsepower, standard speed range	3/4	3/4
Horsepower, with optional speed range	1	1

Requirements are not less than those shown. Where a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

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3.6 Performance. Machine and components, operator's controls and safety devices shall function properly when tested in accordance with 4.5.1.

3.7 Alignment tolerances. Each lathe shall conform to the alignment tolerances shown in table II when tested in accordance with 4.5.2

TABLE II

ALIGNMENT TOLERANCES
(Tolerances in Inches)

TEST	Type I Plain Lathe	Type II Turret Lathe
Bed level longitudinal direction, in 12 inches	0.0005	0.0005
Spindle center run-out	0.0002 TIR	0.0002 TIR
Spindle nose run-out	0.000050 TIR	0.000050 TIR
Cam action of spindle	0.000050 TIR	0.000050 TIR
Headstock and tailstock center alignment, vertical, in 12 inches	0.0005 to 0.001 (high at tailstock)	-
Tailstock spindle alignment horizontal & vertical	0.0002	-
Turret tool hole aligned with spindle		
Horizontal	-	0.0005 TIR
Vertical	-	0.001 TIR
Turret index accuracy	-	± 0.0005
Collet run-out, in 1 inch	0.001 TIR	0.001 TIR
Optional collet run-out accuracy when required (see 6.2)	0.0003 TIR	0.0003 TIR

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3.8 Standard equipment type I lathe. Unless otherwise specified (see 6.2.1), each type I lathe shall be furnished with all standard equipment normally furnished by the machine manufacturer. The equipment specified in 3.8.1 through 3.8.10 shall also be furnished as standard.

3.8.1 Drill chuck. The drill chuck shall be a 0-1/2 inch capacity with taper shank for mounting in the tailstock spindle.

3.8.2 Centers. One headstock center, male, hardened, draw-in collet type and one tailstock center, taper shank, male type, hardened, and one half-center, hardened, shall be furnished.

3.8.3 Drive plate and drive dog. One drive plate fitted to the head-stock spindle and one driving dog shall be furnished.

3.8.4 Round tool post. The tool post shall mount on the compound slide and shall Consist basically of the round, single screw tool post, T-block, tool post collar and rocker.

3.8.5 Square tool post. The tool post shall mount on the compound slide and shall consist basically of a square, open side post and adjustable wedge for centering the cutting edge of the tool. The post shall accept at least a 3/8 inch square tool.

3.8.6 Lever type collet closer. The collet closer shall be the lever operated type mounted on the rear of the spindle. The mechanism shall be quick acting and easily operated, with means for adjusting the work grip tension.

3.8.7 Collets. One set each of fractional collets for round, square and hexagon bars shall be furnished in increments of 1/32 inch to cover the range of sizes specified in table I. When specified, collets for round, square or hexagon bars shall be furnised in 1\64, or 1\16, or 1/8 inch increments, in lieu of 1/32 inch.

3.8.8 Collet stop. One collet stop, adjustable type, for fitting inside the collet shall be furnished. The stop shall be suitable for use with all size collets furnished

3.8.9 Chuck and chuck jaws. The number and type of chuck and chuck jaws shall be as specified (see 6.2.1) , and shall conform to the applicable require-ments of ANSI B5. 8.

3.8.10 Machine light. The machine light shall be for mounting on the lathe. The support arm shall be adjustable for best illumination of the work area. The lamp shall be protected and shall operate on 110 to 125 volts.

3.9 Standard equipment type II lathe. Unless otherwise specified (see 6.2.1), each type II lathe shall be furnished with all standard equipment normally furnished by the machine manufacturer. The equipment specified in 3.9.1 through 3.9.5 shall also be furnished as standard.

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3.9.1 Lever type collet closer. The collet closer shall be the lever operated type mounted on the rear of the spindle. The mechanism shall be quick acting and easily operated, with mans for adjusting the work grip tension.

3.9.2 Collets. One set each of fractional collets for round, square and hexagon bars shall be furnished in increments of 1/32 inch to cover the range of sizes specified in table I. When specified, collets for round, square or hexagon bars shall be furnished in 1/64, or 1/16, or 1/8 inch increments, in lieu of 1/32 inch.

3.9.3 Collet stop. One collet stop, adjustable type, for fitting inside the collet shall be furnished. The stop shall be suitable for use with all size collets furnished.

3.9.4 Chuck and chuck jaws. The number and type of chuck and chuck jaws shall be as specified (6.2.1), and shall conform to the applicable requirements of ANSI B5.8.

3.9.5 Machine light. The machine light shall be for mounting on the lathe. The support arm shall be adjustable for best illumination of the work area. The lamp shall be protected and shall operate on 110 to 125 volts.

3.10 optional equipment. Optional equipment for type lathe required shall be furnished as specified and as fully described (see 6.2.1).

3.11 Additional requirements. Perfomance features, attachments or accessories required in addition to, or in lieu of those specified herein, shall be as required by the procuring activity (see 6.2.1).

3.12 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart if furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means for mounting. The chart or plate shall contain the following;

Points of lubricant application
Service interval
Type of lubricant
Viscosity

3.13 Nameplate: Unless otherwise specified (see 6.2.1), a corrosion-resistant metal nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard lathe and a suffix identified in the manufacturers permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

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Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Type_____, Horsepower_____
Power input (volts, total amps, phase, frequency)
Contract number or order number
National stock number or plant equipment code
Date of manufacture

3.14 Technical data. Technical data shall be furnished as specified (see 6.2.2).

3.15 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. When first article approval is required under 3.1, first article inspection shall be per formed. Unless otherwise specified (see 6.2.1), first article inspection shall comprise the examination in 4.4, and all tests in 4.5. Failure to pass the examination or any test shall be cause for disapproval of the first article.

4.3 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1) , quality conformance inspection shall consist of the examination in 4.4, the test in 4.5.1, and the inspection in 4.6. Failure of an item to pass any examination or test shall be cause for rejection.

4.4 Examination. The machine and equipment shall be examined for compliance with theequipments of this specification in 3.2 throught 3.5 and 3.8 through 3.15.

4.5 Tests.

4.5.1 Performance test. The machine shall be operated in accordance with the builder's standard procedure for warm-up and run-off checks. Proper operation of all controls, motors, adjusting mechanisms and accessories shall be verified during the trial period.

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4.5.2 Alignment accuracy test. The machine shall be tested in accordance with the builder's standard test methods to determine that it complies with the alignment accuracy requirements of 3.7.

4.6 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Packaging - packing and markings. Unless otherwise specified, packaging, packing and marking shall be in accordance with ASTM Standard D 3951. When required, level A packaging and level A or B packing and marking shall be in accordance with the applicable requirements of MIL-M-18058. The required levels and special marking shall be as specified (see 6.2.1).

6. NOTES

6.1 Intended use. The lathes covered by this specification are intended for use in production shops, toolrooms or laboratories to produce small parts or to perform additional operations such as turning, drilling, and boring to close tolerances at high speeds.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Type lathe required (see 1.2).
c. First article approval, if required (see 3.1).
d. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).
e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).
f. Dual calibrations (US and SI), if required (see 3.2.1.2).
g. Additional safety and health requirements (see 3.2.4 and 6.4).
h. Painting, if different (see 3.3.5).
i. Specify threaded type spindle nose, if required (see 3.4.4).
j. Electrical system, if different (see 3.4.5).
k. Type motor enclosure, if different (see 3.4.6).
l Special accuracy collets, if required (see talbe II) .
m. Standard equipment type I lathe, if different (see 3.8) .
n. Specify chuck and chuck jaws required (see 3.8. 9)
o. Standard equipment type II lathe, if different (see 3.9).
p. Specify chuck and chuck jaws required (see 3.4.9).
q. Optional equipment, as required (see 3.10).
r. Additional requirements, as required (see 3.11).
s. Lubrication chart or plate, if different (see 3.12).
t. Nameplate, if different (see 3.13),

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u. First article inspection, if different (see 4.2).
v. Quality fonformance inspection, if different (see 4.3).
w. specific packaging, packing and special marking, if required (see 5.1). (Specify level A packaging, specify level A or B packing and specify special marking, if required.)

6.2.2 Contract data requirements. Required technical data, such as operators' manuals, parts lists, foundation and anchor bolt plans, wiring diagrams and other inspections for operation and maintenance as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract.

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Inquisition Regulation. The first article should be a preproduction sample or a standard production item from the contractor's current inventory. The contracting officer should include specific instructions in all procurement instrument regarding arrangements f& examinations test and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.4 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, heat, etc. Since OSHA limits the total hazard level of those hazards in the environment (and does not limit the hazard level of individual machine in the environment), the requesting activity is advised to analyze the existing hazard levels in the proposed operating environment and specify additional machine requirements that will integrate the new machine into its future operating environment. If specific point-of-operation guarding is required, the requesting activity should specify the exact configuration of the guard required, as in most cases, the guard configuration is dependent on the size and configuration of the workpieces that will be machined. The above and other additional safety and health requirements should be specified in detail under 6.2.1(g).

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes, with respect to the previous. issue, due to the extensiveness of the changes.

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Navy - SH
Air Force - 99

Review Activites:
Air Force - 84
DLA - GS

User Activity:
Navy - MC

Preparing Activity:
DLA - IP

Project Number:
3416-0175

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-8061B	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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